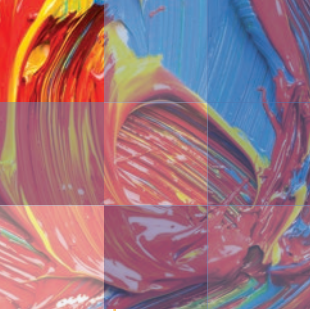




# Systems of Linear Inequalities

A solution of a system of linear inequalities is any ordered pair that satisfies all the inequalities in the system.

## Definition

A **system of linear inequalities** is a set of two or more linear inequalities using the same variables.

Recall that you can solve a system of linear equations by graphing. Similarly, you can also solve a system of linear inequalities by graphing. When two or more linear inequalities are graphed on the same coordinate plane, the solutions of the system are the ordered pairs in the intersection of the shaded regions.

## Solving Systems of Two Linear Inequalities

To solve a system of linear inequalities, graph each inequality in the system and then identify the overlapping region.

### EXAMPLE 1

Graph the system of linear inequalities. Then give one ordered pair that is a solution of the system and check your answer.

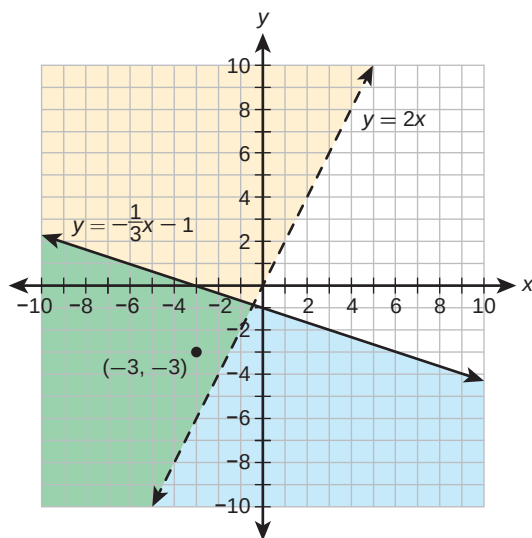
$$\mathbf{A} \quad \begin{cases} y > 2x \\ y \geq -\frac{1}{3}x - 1 \end{cases}$$

### SOLUTION

Graph  $y = 2x$  with a dashed line and shade above the line.

Graph  $y = -\frac{1}{3}x - 1$  with a solid line and shade below it.

The intersection of the half-planes is the green shaded region. The solutions are the ordered pairs in the green region. One solution is the ordered pair  $(-3, -3)$ . You can verify that  $(-3, -3)$  satisfies both inequalities.



### CHECK

Substitute  $(-3, -3)$  in  $y > 2x$ .

$$\begin{aligned}y &> 2x \\ -3 &\stackrel{?}{>} 2(-3) \\ -3 &> -6 \quad \checkmark\end{aligned}$$

Substitute  $(-3, -3)$  in  $y \leq -\frac{1}{3}x - 1$ .

$$\begin{aligned}y &\leq -\frac{1}{3}x - 1 \\ -3 &\stackrel{?}{\leq} -\frac{1}{3}(-3) - 1 \\ -3 &\leq 0 \quad \checkmark\end{aligned}$$

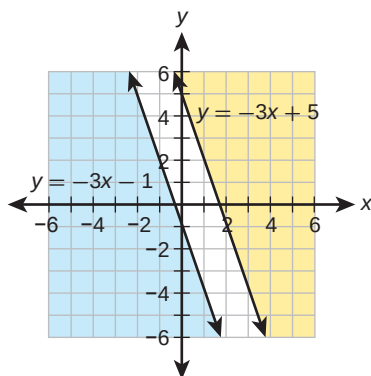
The ordered pair  $(-3, -3)$  satisfies both inequalities, so it is a solution of the system.

► **Think About It** The ordered pair tested is just one of infinitely many solutions.

**B** 
$$\begin{cases} y \geq -3x + 5 \\ y \leq -3x - 1 \end{cases}$$

**SOLUTION**

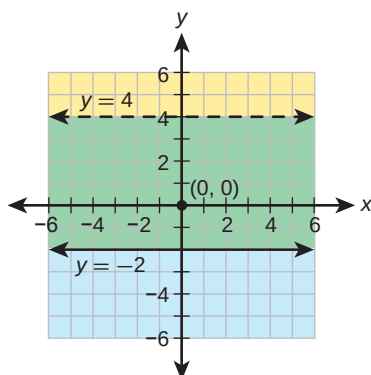
The boundary lines are parallel, and both are graphed with solid lines. One is shaded above the line, and the other is shaded below. The shaded areas do not overlap; therefore, this system of linear inequalities has no solution.



**C** 
$$\begin{cases} y < 4 \\ y \geq -2 \end{cases}$$

**SOLUTION**

The boundary lines are parallel, but the shading is in opposite directions, creating an overlapping region. This region contains the solutions of the system. One solution is  $(0, 0)$ .



# Solving Systems of More Than Two Linear Inequalities

When a system has three or more inequalities, graph each inequality on the same coordinate plane. Be careful to identify the overlapping regions correctly.

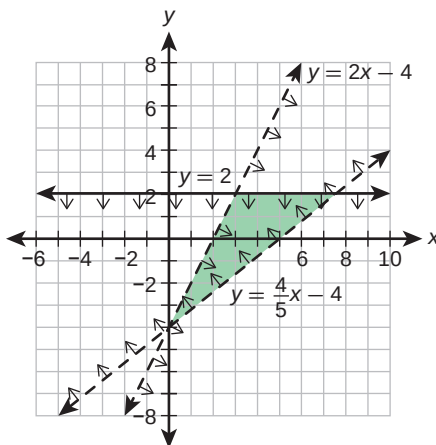
## EXAMPLE 2

Graph the system of linear inequalities. Identify the overlapping shaded regions representing the solution of the system.

**A** 
$$\begin{cases} y < 2x - 4 \\ y \leq 2 \\ y > \frac{4}{5}x - 4 \end{cases}$$

### SOLUTION

Graph  $y = 2x - 4$  with a dashed line and use small arrows below the line to show which half-plane should be shaded. Graph  $y = 2$  with a solid line and draw small arrows below the line. Graph  $y = \frac{4}{5}x - 4$  with a dashed line and draw small arrows above the line. The solution is the shaded triangular region.



**B**

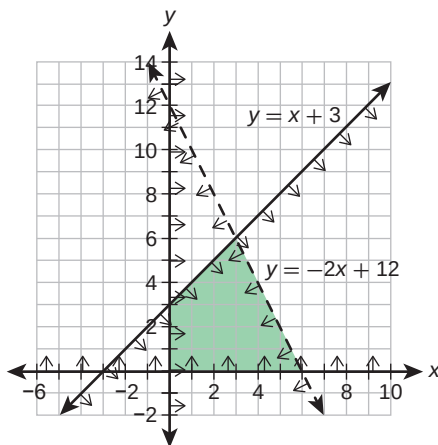
$$\begin{cases} x \geq 0 \\ y \geq 0 \\ y \leq x + 3 \\ y < -2x + 12 \end{cases}$$

**SOLUTION**

The first two inequalities,  $x \geq 0$  and  $y \geq 0$ , describe Quadrant I of a coordinate graph and the adjoining sections of the  $x$ - and  $y$ -axes.

Graph the line  $y = x + 3$  with a solid line and draw small arrows below the line. Graph the line  $y = -2x + 12$  with a dashed line and draw small arrows below the line.

The solutions of the system are found in the region bounded by the  $x$ - and  $y$ -axes and the lines  $y = x + 3$  and  $y = -2x + 12$ . Note that  $(0, 0)$  and  $(0, 3)$  are solutions of the system because they represent the intersections of solid lines. However,  $(6, 0)$  and  $(3, 6)$  are not solutions of the system because they represent the intersections of solid lines and dashed lines.

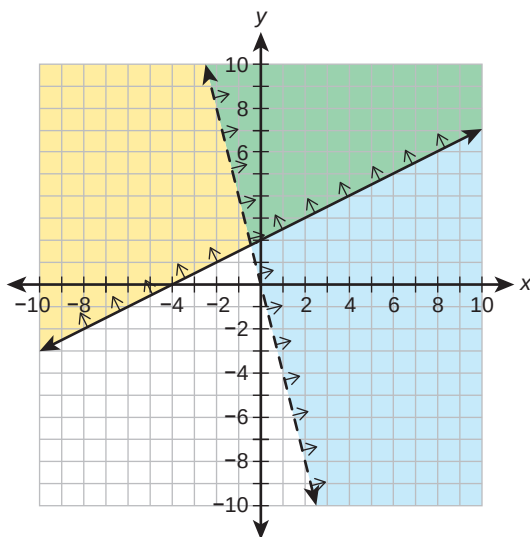


# Writing Systems of Linear Inequalities from Graphs

You can write a system of linear inequalities from a graph by determining the equation for each boundary line.

## EXAMPLE 3

Write the system of inequalities shown by the graph.



## SOLUTION

**Step 1** Determine the equations of the boundary lines.

The  $y$ -intercept of the solid line is 2 and its slope is  $\frac{1}{2}$ . The equation of the boundary line is  $y = \frac{1}{2}x + 2$ .

The  $y$ -intercept of the dashed line is  $-1$  and its slope is  $-4$ . The equation of the boundary line is  $y = -4x - 1$ .

**Step 2** Determine the inequality signs.

Because the boundary line of  $y = \frac{1}{2}x + 2$  is solid, the inequality will be a nonstrict inequality, using either  $\leq$  or  $\geq$ . The shading is above the line, so use  $\geq$ .

Because the boundary line of  $y = -4x - 1$  is dashed, the inequality will be a strict inequality, using  $<$  or  $>$ . The shading is above the line, so use  $>$ .

**Step 3** Write the system of linear inequalities.

$$\begin{cases} y \geq \frac{1}{2}x + 2 \\ y > -4x - 1 \end{cases} \quad \blacksquare$$